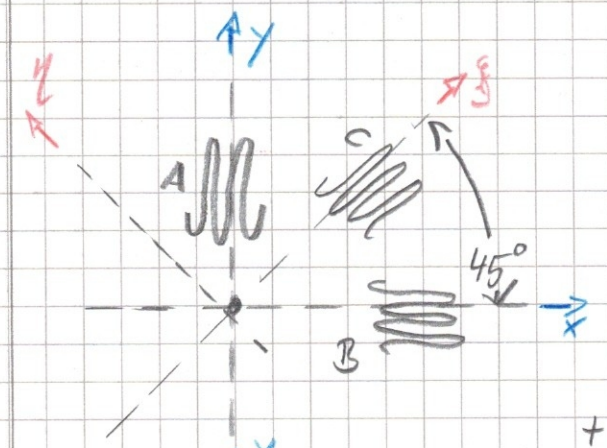
Exercise strain gauge

LMMD
20.04.26
WUTTKE

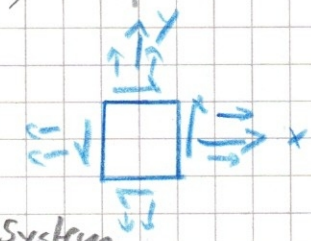


$\varepsilon_A = \varepsilon_y$

$\varepsilon_B = \varepsilon_x$

$\varepsilon_C = \varepsilon_\xi$

+ plane stress $\Rightarrow \sigma_z = 0$
(unloaded surface) $\tau_{yz} = \tau_{xz} = 0$

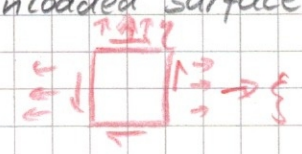


x-y-System

(1) $E \cdot \underline{\varepsilon_x} = \sigma_x \rightarrow \sigma_y$

(2) $E \cdot \underline{\varepsilon_y} = \sigma_y \rightarrow \sigma_x$

$G \cdot \gamma_{xy} = \tau_{xy}$

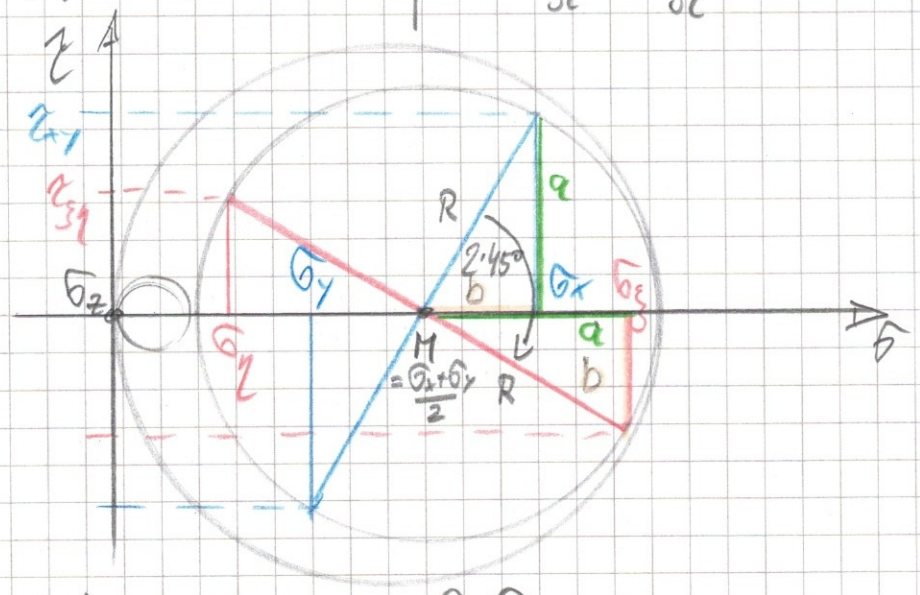


ξ-η-system

(3) $E \cdot \underline{\varepsilon_\xi} = \sigma_\xi \rightarrow \sigma_\eta$

$E \cdot \varepsilon_\eta = \sigma_\eta \rightarrow \sigma_\xi$

$G\, \gamma_{\xi\eta} = \tau_{\xi\eta}$



$$\Rightarrow \sigma_\xi = M + a = \frac{\sigma_x + \sigma_y}{2} + \tau_{xy} \quad (4)$$

$$\sigma_\eta = M - a = \frac{\sigma_x + \sigma_y}{2} - \tau_{xy} \quad (5)$$

$\Rightarrow$ equation (4) and (5) in (3) $\Rightarrow E\varepsilon_\xi = f(\sigma_x; \sigma_y; \tau_{xy})$
with (1) and (2) now we have three equations for the three unknowns σ_x σ_y τ_{xy}.

a) (4) and (5) in (3):

$$E \cdot \varepsilon_\xi = \frac{\sigma_x + \sigma_y}{2} + \tau_{xy} - \nu \left[\frac{\sigma_x + \sigma_y}{2} - \tau_{xy} \right]$$

$$(6) \quad E \cdot \varepsilon_\xi = \frac{\sigma_x}{2}(1-\nu) + \frac{\sigma_y}{2}(1-\nu) + \tau_{xy}(1+\nu)$$

b) with (1) and (2):

$$(6) \quad 2E\varepsilon_\xi = \sigma_x - \nu\sigma_x + \sigma_y - \nu\sigma_y + 2\tau_{xy}(1+\nu)$$

$$(1) \quad E\varepsilon_x = \sigma_x - \nu\sigma_y$$

$$(2) \quad E\varepsilon_y = -\nu\sigma_x + \sigma_y$$

$$(6)-(1)-(2) \quad 2E\varepsilon_\xi - E\varepsilon_x - E\varepsilon_y = 2\tau_{xy}(1+\nu)$$

$$\Rightarrow \tau_{xy} = E \frac{2\varepsilon_\xi - \varepsilon_x - \varepsilon_y}{2 \cdot (1+\nu)}$$

with (1) and (2) direct:

$$\begin{cases} \sigma_x = E \dfrac{\varepsilon_x + \nu\varepsilon_y}{1-\nu^2} \\ \sigma_y = E \dfrac{\varepsilon_y + \nu\varepsilon_x}{1-\nu^2} \end{cases}$$

here:

$$\left.\begin{aligned} \varepsilon_A &= \varepsilon_y = 0{,}2 \cdot 10^{-4} \\ \varepsilon_B &= \varepsilon_x = 5{,}8 \cdot 10^{-4} \\ \varepsilon_C &= \varepsilon_\xi = 4{,}3 \cdot 10^{-4} \end{aligned}\right\} \Rightarrow \left\{\begin{aligned} \sigma_x &= 121{,}7 \text{ MPa} \\ \sigma_y &= 28{,}33 \text{ MPa} \\ \tau_{xy} &= 21{,}67 \text{ MPa} \end{aligned}\right.$$

$E = 200\,000$ MPa

$\nu = 0{,}2$